

RAJA SATYA VARDHANA REDDY THIPPAREDDY

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ASSOCIATE AI ENGINEER | PYTHON | APPLIED AI | BACKEND INTEGRATION

PROFESSIONAL SUMMARY

Early-career AI engineer with graduate training in Artificial Intelligence, Machine Learning, and IoT and hands-on experience building multimodal AI, deep-learning, local LLM, healthcare decision-support, and connected-device systems. Strong foundation in Python, machine learning, REST APIs, SQL, model evaluation, debugging, and AI-service integration, with practical experience evaluating 500+ LLM and multimodal outputs. Comfortable learning unfamiliar technologies quickly, translating requirements into working prototypes, and collaborating across technical and non-technical teams. Focused on building reliable AI-powered features, backend services, and intelligent workflows that solve real business problems.

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript

AI & Machine Learning: Scikit-learn, TensorFlow, PyTorch, CNN, RNN, LSTM, InceptionV3, Transfer Learning, Computer Vision, Model Evaluation

Generative AI: GPT APIs, Google Vision API, Ollama, Local LLMs, Prompt Engineering, LLM Evaluation, Quantization, RAG Concepts

Backend & Integration: FastAPI, REST APIs, JSON, API Integration, Microservice Fundamentals, Relational Database Fundamentals

Data & Cloud: Pandas, NumPy, Data Preprocessing, ETL Concepts, Data Pipelines, Feature Engineering, Google Cloud Platform, Vertex AI, AWS Foundations

Engineering: Git, GitHub, Testing, Debugging, Technical Documentation, Code Review Concepts, Agile Fundamentals, CI/CD and MLOps Concepts

PROFESSIONAL EXPERIENCE

AI Evaluator & Performance Analyst - Outlier AI

Remote | Nov 2025 - Apr 2026

- Evaluated 500+ LLM and multimodal outputs for factual accuracy, reasoning quality, instruction adherence, safety, clarity, and consistency using structured evaluation criteria.
- Identified hallucinations, weak reasoning, missed constraints, and inconsistent model behaviour, documenting failure patterns and recommendations to support reliable AI output.
- Worked across prompt evaluation, multimodal annotation, classification, and benchmarking-style tasks, strengthening practical understanding of LLM behaviour, model quality, and AI-system evaluation.

Crew Trainer - Tim Hortons

Ontario, Canada | Jan 2024 - Aug 2025

- Trained team members and adapted quickly to changing workflows in a high-volume environment while maintaining process accuracy and service quality.
- Built strong communication, teamwork, prioritization, and problem-solving skills by resolving operational issues under time pressure.

EDUCATION

Graduate Certificate - Artificial Intelligence with Machine Learning | Humber Polytechnic, Canada | 2025

Graduate Certificate - Internet of Things with Machine Learning | Sheridan College, Canada | 2024

Bachelor of Technology - Information Technology | Vellore Institute of Technology, India | 2023

SELECTED AI & ENGINEERING PROJECTS

AR Vision Assistant - Multimodal AI System

Humber Polytechnic Capstone | Python, Google Vision API, GPT APIs, Multimodal AI, API Integration

- Built a visual-to-response AI pipeline that processed image input, extracted contextual information using Google Vision API, and generated user-facing interpretations through GPT APIs.
- Refined prompts, validated AI outputs, troubleshooted API behaviour, and documented system architecture and integration decisions throughout prototype development.
- Led technical presentation activities, translating system architecture, implementation trade-offs, and prototype outcomes for sponsor and faculty stakeholders.

MERLIN - Medical Emergency Routing and Live Intelligence Network

Independent Healthcare AI Project | Python, Simulation, Dispatch Logic, Decision Support, Evaluation

- Developed a Python-based emergency-response simulation comparing ambulance and helicopter dispatch decisions using vehicle availability, patient severity, routing, and cost-time trade-offs.
- Implemented and compared random dispatch, nearest-vehicle assignment, and severity-rule baselines to evaluate whether the proposed decision-support approach provided measurable value.
- Structured simulation outputs and visualizations to make system behaviour, trade-offs, and evaluation results easier to interpret and reproduce.

Offline LLM Deployment & Quantization

Academic Project | Ollama, Local LLMs, Quantization, Offline Inference

- Deployed local LLM environments using Ollama to explore AI inference without depending entirely on cloud-hosted model services.
- Compared model configuration and quantization trade-offs involving compute requirements, inference efficiency, local performance, and deployment constraints.

Sign Language Recognition System

VIT B.Tech Final-Year Capstone | Python, TensorFlow, CNN, RNN, LSTM, InceptionV3

- Developed a video-based sign-language recognition prototype using CNN/RNN/LSTM architectures and InceptionV3 transfer learning for gesture interpretation.
- Explored sequential gesture patterns and model architectures while applying training, evaluation, debugging, and computer-vision techniques to an accessibility-focused problem.

Visitor Monitoring & Smart Door System

Sheridan College IoT Capstone | ESP32, Ultrasonic Sensors, Blynk, Embedded Systems

- Designed an IoT smart-access prototype using ESP32 microcontrollers and ultrasonic sensors for visitor monitoring and gate-control functionality.
- Integrated Blynk for remote monitoring and improved reliability through sensor calibration, repeated testing, debugging, and hardware-software integration.

ADDITIONAL INFORMATION

Work Authorization: Canadian Open Work Permit | Authorized for full-time employment | No employer sponsorship required
Location & Availability: Hamilton, Ontario | Available for ShyftLabs' Toronto hybrid work model